Problem 5

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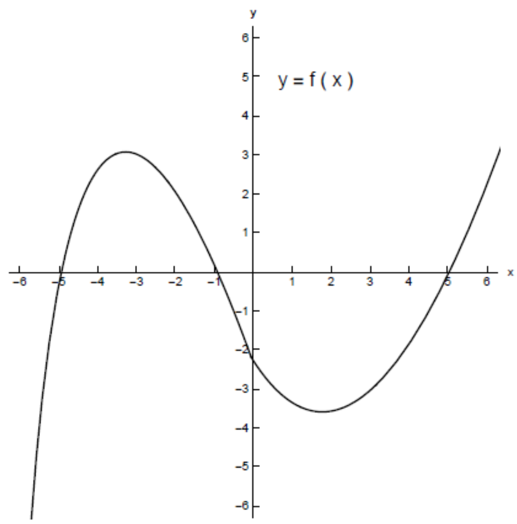
Problem 1 *Suppose we are given the graph of a function f:*



(a) *Use this graph to find the following: (Assume all values will be integers or $+\infty$ or $-\infty$)*

  (i) *all x where $f(x) = 0$,*

  (ii) *all x where $f(x) > 0$,*

  (iii) *all x where $f(x) < 0$, and*

  (iv) *all x where $f(x)$ attains a local maximum and all x where f attains a local minimum.*

Without sketching the graph of f' find

(b) (i) *all x where $f'(x) = 0$,*

  (ii) *all x where $f'(x) > 0$,*

  (iii) *all x where $f'(x) < 0$, and*

  (iv) *On the following intervals, does $f'(x)$ seem to be increasing or decreasing?*

    i. $(-\infty, 0)$

    ii. $(0, \infty)$

(c) *Sketch a graph of f'.*
[2]